

Hands-on exercises 8: Degenerate electron gas and properties of white dwarfs

P. Käpylä, I. Milić

July 10, 2025

Problem 1: Use the Heisenberg uncertainty principle and Pauli exclusion principle to derive the equations of state for non-relativistic and relativistic degenerate electron gas:

$$p_{\text{e,deg}} = \frac{h^2}{20m_e} \left(\frac{3}{\pi}\right)^{2/3} \frac{1}{m_{\text{H}}^{5/3}} \left(\frac{\rho}{\mu_e}\right)^{5/3}, \quad (1)$$

and

$$p_{\text{e,r-deg}} = \frac{hc}{8} \left(\frac{3}{\pi}\right)^{1/3} \frac{1}{m_{\text{H}}^{4/3}} \left(\frac{\rho}{\mu_e}\right)^{4/3}, \quad (2)$$

where $h = 6.626 \cdot 10^{-34} \text{ J Hz}^{-1}$ is the Planck constant, m_e is the mass of the electron, m_{H} is the atomic mass unit, and μ_e^{-1} is the average number of free electron per nucleon.

Compare the degenerate electron pressure and the the gas pressure from ideal gas equation at the centre of the Sun with solar composition of $\mu = 0.62$ and $\mu_e = 1.17$. Use $\rho_c = 1.6 \cdot 10^5 \text{ kg m}^{-3}$ and $T_c = 1.57 \cdot 10^7 \text{ K}$ for the central density and temperature of the Sun.

Solution: The Heisenberg uncertainty principle states that the position and momentum (p_{m}) of a particle cannot be determined arbitrarily accurately. Quantitatively this is given by:

$$\Delta V \Delta^3 p_{\text{m}} \geq h^3, \quad (3)$$

where h is the Planck constant. On the other hand, according to the Pauli exclusion principle no two particles can occupy the same quantum state (same momentum and spin). Therefore, in a degenerate electron gas an element in phase space (location and momentum) can be occupied by two electrons (spin up and down). In a state of complete degeneration all the quantum states up to a maximum momentum value are occupied and Eq. (3) becomes equality. Then the number of electrons with momenta between $(p_{\text{m}}, p_{\text{m}} + dp_{\text{m}})$ per unit volume:

$$n_{\text{e}}(p_{\text{m}})dp_{\text{m}} = \frac{2}{\Delta V} = \frac{2}{h^3} 4\pi p_{\text{m}}^2 dp_{\text{m}}. \quad (4)$$

Integrating this equation yields

$$\int_0^{p_{\text{m}0}} n_{\text{e}}(p_{\text{m}})dp_{\text{m}} = \int_0^{p_{\text{m}0}} \frac{2}{h^3} 4\pi p_{\text{m}}^2 dp_{\text{m}}, \quad (5)$$

$$n_{\text{e}} = \frac{8\pi}{3h^3} p_{\text{m}0}^3, \quad \text{or} \quad p_{\text{m}0} = \left(\frac{3n_{\text{e}}h^3}{8\pi}\right)^{1/3}. \quad (6)$$

Now we use the pressure integral from the lectures:

$$p = \frac{1}{3} \int u p_m n(p_m) dp_m, \quad (7)$$

where we now use $u = p_m/m_e$ and $n = n_e$:

$$p_{e,\text{deg}} = \frac{1}{3} \int_0^{p_{m0}} \frac{p_m^2}{m_e} n_e(p_m) dp_m. \quad (8)$$

Substituting n_e from Eq.(4) and integrating gives:

$$p_{e,\text{deg}} = \frac{1}{3} \int_0^{p_{m0}} \frac{p_m^2}{m_e} \frac{2}{h^3} 4\pi p_m^2 dp_m. \quad (9)$$

$$= \frac{8\pi}{15m_e h^3} p_{m0}^5. \quad (10)$$

Now we make use of Eq.(6) and recast the electron number density in terms of the gas density (lectures)

$$n_e = \frac{\rho}{\mu_e m_H}. \quad (11)$$

Substituting these in Eq. (10) gives

$$p_{e,\text{deg}} = \frac{8\pi}{15m_e h^3} \left(\frac{3n_e h^3}{8\pi} \right)^{5/3}, \quad (12)$$

$$= \frac{8\pi}{15m_e h^3} \left(\frac{3h^3}{8\pi} \frac{\rho}{\mu_e m_H} \right)^{5/3}. \quad (13)$$

Rearranging this yields the desired formula:

$$p_{e,\text{deg}} = \frac{h^2}{20m_e} \left(\frac{3}{\pi} \right)^{2/3} \frac{1}{m_H^{5/3}} \left(\frac{\rho}{\mu_e} \right)^{5/3} \equiv K_1 \rho^{5/3}. \quad (14)$$

In the (ultra-)relativistic case the the velocity of the electrons approaches the speed of light. Then we replace u in the pressure integral (7) by c

$$p_{e,\text{r-deg}} = \frac{1}{3} \int_0^{p_{m0}} c p_m n_e(p_m) dp_m. \quad (15)$$

Substituting (4) and integrating yields:

$$p_{e,\text{r-deg}} = \frac{1}{3} \int_0^{p_{m0}} c p_m \frac{2}{h^3} 4\pi p_m^2 dp_m. \quad (16)$$

$$= \frac{2\pi c}{3h^3} p_{m0}^4. \quad (17)$$

We again make use of Eqs.(6) and (11):

$$p_{e,\text{r-deg}} = \frac{2\pi c}{3h^3} \left(\frac{3n_e h^3}{8\pi} \right)^{4/3}, \quad (18)$$

$$= \frac{2\pi c}{3h^3} \left(\frac{3h^3}{8\pi} \frac{\rho}{\mu_e m_H} \right)^{4/3}. \quad (19)$$

Rearranging this yields the desired formula:

$$p_{\text{e,r-deg}} = \frac{hc}{8} \left(\frac{3}{\pi} \right)^{1/3} \frac{1}{m_{\text{H}}^{4/3}} \left(\frac{\rho}{\mu_{\text{e}}} \right)^{4/3} \equiv K_2 \rho^{4/3}. \quad (20)$$

To compare the gas pressure and degenerate electron pressure in the Sun, we compute the ratio $p_{\text{e,deg}}/p_{\text{gas}}$, where

$$p_{\text{gas}} = \frac{\mathcal{R}}{\mu} \rho T, \quad (21)$$

and where the specific gas constant is given by $\mathcal{R} = k/m_{\text{H}}$,

$$\frac{p_{\text{e,deg}}}{p_{\text{gas}}} = \frac{h^2}{20m_{\text{e}}} \left(\frac{3}{\pi} \right)^{2/3} \frac{1}{m_{\text{H}}^{5/3}} \frac{\mu}{\mu_{\text{e}}^{5/3}} \frac{\rho^{2/3}}{\mathcal{R}T}. \quad (22)$$

Inserting the numerical values yields

$$\frac{p_{\text{e,deg}}}{p_{\text{gas}}} \approx 0.11, \quad (23)$$

which means that electron degeneracy is beginning to have an effect in the solar centre, although gas pressure still dominates.

Problem 2: Derive the mass-radius relationship of white dwarfs assuming non-relativistic degenerate electron gas and the Chandrasekhar mass using the Lane-Emden equation. Hint: recall the for the equation of state of non-relativistic (relativistic) degenerate electron gas the polytropic index is $n = \frac{3}{2}$ ($n = 3$), and assume that hydrogen has been depleted so that $\mu_{\text{e}} = 2$.

Solution: Let us first recall the Lane-Emden equation:

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\theta^n. \quad (24)$$

where $\xi = \alpha r$ and $\rho = \rho_{\text{c}} \theta^n$ where n is the polytropic index related to the adiabatic index via

$$\gamma = 1 + \frac{1}{n}. \quad (25)$$

The mass of the star is given by:

$$M = \int_0^R 4\pi r^2 \rho dr. \quad (26)$$

We now make a substitution of variables $r = \xi/\alpha$, $dr = d\xi/\alpha$, and $\rho = \rho_{\text{c}} \theta^n$, and set the integration limits to $(0, \xi_{\text{s}})$ where ξ_{s} is the surface corresponding to the first zero-crossing of the solution of Eq. (24). We obtain:

$$M = \frac{4\pi\rho_{\text{c}}}{\alpha^3} \int_0^{\xi_{\text{s}}} \xi^2 \theta^n d\xi. \quad (27)$$

Now we substitute the Lane-Emden equation, Eq. (24), to find:

$$M = -\frac{4\pi\rho_{\text{c}}}{\alpha^3} \int_0^{\xi_{\text{s}}} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) d\xi. \quad (28)$$

We may also write $\xi_s = \alpha R$, where R is the radius of the star to eliminate α . Furthermore, it is easy to integrate Eq. (28) bearing in mind that $d\theta/d\xi = 0$ at $\xi = 0$:

$$M = -4\pi\rho_c \left(\frac{R}{\xi_s}\right)^3 \int_0^{\xi_s} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi}\right) d\xi = -4\pi\rho_c \left(\frac{R}{\xi_s}\right)^3 \left(\xi_s^2 \frac{d\theta}{d\xi}\bigg|_{\xi_s}\right) = 4\pi\rho_c R^3 \left(-\frac{1}{\xi_s} \frac{d\theta}{d\xi}\bigg|_{\xi_s}\right). \quad (29)$$

Now we recall from the homework that

$$\alpha^2 = \frac{4\pi G}{K(n+1)} \rho_c^{1-\frac{1}{n}}, \quad \text{and} \quad p = K\rho^{1-\frac{1}{n}}. \quad (30)$$

We eliminate α with the definition of ξ_s and ρ_c using Eq. (29):

$$\frac{\xi_s^2}{R^2} = \frac{4\pi G}{K(n+1)} \left[\frac{M}{4\pi R^3 \left(-\frac{1}{\xi_s} \frac{d\theta}{d\xi}\bigg|_{\xi_s}\right)} \right]^{1-\frac{1}{n}}. \quad (31)$$

Assuming that we have solved the Lane-Emden equation we know ξ_s and $-\frac{1}{\xi_s} \frac{d\theta}{d\xi}\bigg|_{\xi_s}$ and identifying that everything except M and R are constants, we find:

$$\begin{aligned} R^{-2} &\propto \frac{M^{1-\frac{1}{n}}}{R^{3-\frac{3}{n}}}, \\ R &\propto M^{\frac{n-1}{n-3}}. \end{aligned} \quad (32)$$

Inserting $n = \frac{3}{2}$ for non-relativistic electron gas gives

$$R \propto M^{-1/3}, \quad (33)$$

which shows that the radius of the white dwarf decreases with mass. If we use $n = 3$ corresponding to relativistic electron gas, we see that the radius approaches zero suggesting that something interesting happens at this limit.

Going back to Eq. (31) and substituting $n = 3$, we find that R drops out from the equation. Furthermore, solving for M yields

$$M = 4\pi \left(\frac{K}{\pi G}\right)^{3/2} \left(-\xi_s^2 \frac{d\theta}{d\xi}\bigg|_{\xi_s}\right). \quad (34)$$

Identifying that $K = K_2$ from Eq. (20) and that for a polytrope of $n = 3$, $-\xi_s^2 \frac{d\theta}{d\xi}\bigg|_{\xi_s} = 2.02$, we get

$$M \approx 2.89 \cdot 10^{30} \text{ kg} = 1.45 M_\odot \equiv M_{\text{Ch}}. \quad (35)$$

Useful physical constants

- $R_\odot = 696 \times 10^6 \text{ m}$
- $M_\odot = 1.989 \times 10^{30} \text{ kg}$

-
- $L_{\odot} = 3.83 \times 10^{26} \text{ W}$
 - $T_{\odot}^{\text{eff}} = 5777 \text{ K}$
 - $1 \text{ AU} = 1.496 \times 10^8 \text{ km}$
 - $c = 2.997 \times 10^8 \text{ m/s}$
 - $G = 6.674 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
 - $k = 1.38 \cdot 10^{-23} \text{ J/K}$
 - $m_e = 9.11 \cdot 10^{-31} \text{ kg}$
 - $m_H = 1.67 \cdot 10^{-27} \text{ kg}$
-